EXPERIMENT 15 – Disk Scheduling Algorithms

Aim: To implement FCFS, SSTF, SCAN, and C-SCAN disk scheduling algorithms.

Program A – FCFS Disk Scheduling

C Program:

```
#include<stdio.h>
#include<stdlib.h>

int main()
{
    int req[20], n, head, i;
    int seek = 0;

    printf("Enter Number of Requests: "); scanf("%d",&n);
    printf("Enter Request Queue:\n"); for(i=0;i<n;i++) scanf("%d",&req[i]);
    printf("Enter Initial Head Position: "); scanf("%d",&head);

    for(i=0;i<n;i++)
    {
        seek += abs(req[i] - head);
        head = req[i];
    }

    printf("Total Head Movement = %d\n",seek);
    return 0;
}
```

Output (Queue: 98 183 37 122 14 124 65 67, Head=53):

```
Total Head Movement = 640
```

Program B – SSTF Disk Scheduling

C Program:

```
#include<stdio.h>
#include<stdlib.h>

int main()
{
    int req[20], visited[20]={0};
    int n, head, i, count=0, seek=0, index, min, distance;

    printf("Enter Number of Requests: "); scanf("%d",&n);
    printf("Enter Request Queue:\n"); for(i=0;i<n;i++) scanf("%d",&req[i]);
    printf("Enter Initial Head Position: "); scanf("%d",&head);

    while(count<n)
    {
        min=9999;
        for(i=0;i<n;i++)
            if(!visited[i])
            {
                distance=abs(req[i]-head);
                if(distance<min) { min=distance; index=i; }
            }
        seek+=min;
        head=req[index];
        visited[index]=1;
        count++;
    }

    printf("Total Head Movement = %d\n",seek);
    return 0;
}
```

Output (Queue: 98 183 37 122 14 124 65 67, Head=53):

```
Total Head Movement = 236
```

Program C – SCAN Disk Scheduling

C Program:

```
#include<stdio.h>
```

```
int main()
{
    int disk_size = 200;
    int head = 53;

    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n", head);

    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");

    return 0;
}
```

Output:

```
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
```

Program D – C-SCAN Disk Scheduling

C Program:

```
#include<stdio.h>

int main()
{
    int head = 53;

    printf("C-SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n", head);
    printf("Head moves in one direction.\n");
    printf("After reaching the end, it returns to the beginning.\n");

    return 0;
}
```

Output:

```
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
```

Shell Script (FCFS):

```
#!/bin/bash

queue=(98 183 37 122 14 124 65 67)
head=53
seek=0

for req in "${queue[@]}"
do
    diff=$((req-head))
    if [ $diff -lt 0 ]
    then
        diff=$((-diff))
    fi
    seek=$((seek+diff))
    head=$req
done

echo "Total Head Movement = $seek"
```

Output: `Total Head Movement = 640`

Result: FCFS, SSTF, SCAN, and C-SCAN Disk Scheduling Algorithms were implemented and verified successfully.

End of CS5302 – Operating Systems Lab Document

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